

AMENDMENT NO. 1 APRIL 2025
TO
IS 17117 : 2019 HAIR SHAMPOO FOR BABIES — SPECIFICATION

[Page 2, Table 1] — Substitute the existing Table 1 with the below table.

Sl No.	Characteristic(s)	Requirement	Method of Test, Ref to Annex of IS 17117/IS
(1)	(2)	(3)	(4)
i)	Active surfactant content, percent by mass, <i>Min</i> (either of the following to be referred)	2 percent	
a)	Active detergent content, as SLES		Annex A
b)	Active non-ionic and/or anionic detergent content		Annex B
c)	Milder surfactants - Glycinates and/or betaines		Annex H - Method for glycinates and betaines as per annex given
ii)	Non-volatile alcohol soluble matter, percent by mass, <i>Min</i>	4	Annex C
iii)	pH at (27 ± 2) °C	4.0 to 7.5	Annex D
iv)	Foam height of two percent solution, mm, <i>Min</i>	100	Annex E
v)	Heavy metals (as Pb), parts per million, <i>Max</i>	20	Annex F
vi)	Arsenic (as As ₂ O ₃), parts per million, <i>Max</i>	2	Annex G or IS 16913 or IS 17495
vii)	Microbial content/limit:		
	Total microbial count, CFU/ml, <i>Max</i>	100	IS 14648
a)	Yeast and mould count, CFU/ml, <i>Max</i>	100	IS 14648
b)	<i>Escherichia coli</i> , per gram or per ml	Absent	IS 14648
c)	<i>Pseudomonas aeruginosa</i> , per gram or per ml	Absent	IS 14648
d)	<i>Staphylococcus aureus</i> , per gram or per ml	Absent	IS 14648
e)	<i>Candida albicans</i> , per g or per ml	Absent	IS 14648
viii)	Formaldehyde ^{a)} , <i>Max</i> ppm	10	Annex J (UV-vis) or Annex K (HPLC) or Annex M (GC HS)

NOTE — For formaldehyde content UV-Vis method will be considered referee method in case of dispute, unless otherwise justified.⁷

Amendment No. 1 to IS 17117 : 2019

(Page 10, Annex G) — Insert the following new annexes at the end:

‘ANNEX H

[Table 1]

**DETERMINATION OF MILD SURFACTANTS (GLYCINATES AND BETAINES) FROM
BABY SHAMPOO**

H-1 GENERAL

This method is suitable for the quantification of alkyl amidopropyl betaine and alkyl glycinate in baby shampoo using reversed-phase HPLC with UV detection. The samples are dissolved in methanol, filtered and analyzed by reversed-phase HPLC using a polar-embedded C18 column. The concentration of surfactant is determined using a UV detector and a three-point calibration curve. Typically, only the response from the C12 component peak of the surfactant is used for quantifying the analyte of interest.

H-2 APPARATUS

H-2.1 Analytical Balance — with precision of ± 0.05 mg

H-2.2 Magnetic Stirrer

H-2.3 Volumetric Flasks — 50 ml and 100 ml

H-2.4 Volumetric Pipets — Class A, 1 ml, 5 ml and 10 ml

H-2.5 Beaker — 1 500 ml

H-2.6 Graduated Cylinder — 1 000 ml

H-2.7 PTFE syringe filters — 0.45 μ m, (waters corp P/N WAT200500 or equivalent)

H-2.8 Nylon Solvent Filters — 0.45 μ m, (waters corp P/N WAT200532 or equivalent)

H-3 REAGENTS

H-3.1 Acetonitrile — HPLC grade (**CAUTION:** Harmful, flammable)

H-3.2 Methanol — HPLC grade (**CAUTION:** Toxic, flammable)

H-3.3 Phosphoric acid — 85 percent (**CAUTION:** Corrosive)

H-3.4 Raw material reference standard (prefer to use the one added in the formulation. Manufacturer can be contacted to provide respective reference standard).

Examples of glycinate and betaine commonly used are mentioned below.

- a) Sodium cocoyl glycinate - CAS 90387-74-9; and
- b) Coco amido propyl betaine - CAS 61789-40-0.

H-3.5 Water — HPLC grade or better

H-4 PREPARATION OF 0.1 PERCENT PHOSPHORIC ACID MOBILE PHASE

H-4.1 Using a 1 000 ml graduated cylinder, measure 1 000 ml of water and transfer to a mobile phase bottle.

H-4.2 Using a Class A pipette, transfer 1 ml of phosphoric acid to the mobile phase bottle.

H-4.3 Mix well and filter through a nylon solvent filter.

Amendment No. 1 to IS 17117 : 2019

H-5 PRIMARY STANDARD PREPARATION (1.0 mg/ml)

H-5.1 Correcting for purity, weigh (0.10 ± 0.01) g of each surfactant into a 100 ml volumetric flask.

H-5.2 Dilute to volume with methanol and mix well.

H-6 WORKING STANDARDS (0.04 mg/ml, 0.20 mg/ml and 0.40 mg/ml)

H-6.1 Pipet 1 ml, 5 ml and 10 ml aliquots of the primary standard (**H-5**) into three separate 25 ml volumetric flasks.

NOTE — Alternatively, only one standard to be prepared, that is, 0.20 mg/ml. The surfactant calculation can be done against the average area of replicate standards.

H-6.2 Dilute each flask to volume with methanol and mix well.

H-6.3 Filter each standard solution through a 0.45 μ m PTFE syringe filter into an autosampler vial.

H-7 SAMPLE PREPARATION (0.7 PERCENT TO 6.7 PERCENT SURFACTANT)

H-7.1 Weigh (0.15 ± 0.02) g of sample into a 25 ml volumetric flask.

NOTE — Sample weight can be adjusted based on the concentration of surfactant used in the formula to match the concentration of standard and sample.

H-7.2 Add approximately 15 ml of methanol and mix using a stir bar for a minimum of 15 min to disperse the sample.

H-7.3 Dilute sample to volume with methanol.

H-7.4 Filter the sample solution through a 0.45 μ m PTFE syringe filter into an autosampler vial.

H-8 PROCEDURE

H-8.1 5 Replicate Standard Injection RSD NMT 2 Percent

Check standard (after 10 injection or end of sequence run) area difference should NMT 3 percent respect to calibration curve respective standard area or average area of 5 replicate standard in system suitability.

H-8.2 Inject each sample with an appropriate number of mobile phase and/or diluent blanks onto the HPLC.

H-8.3 Inject the mid-point calibration standard onto the HPLC at least once every 10 sample injections and again at the end of the sequence run.

H-8.4 Prepare a calibration curve by plotting peak area of the C12 component of the respective surfactant versus concentration in mg/ml of the working standard solutions using a linear regression, which does not force the calibration line through zero.

NOTE — Alternatively, only one standard to be prepared, that is, 0.20 mg/ml. The surfactant calculation can be done against the average area of replicate standards.

H-9 HPLC CONDITIONS

Instrument	:	Chromatograph equipped with an autosampler and UV detector or equivalent
Column	:	Waters Xbridge C18, (4.6 x 150) mm, 3.5 μ m or YMC pack ODS-A (150 x 4.6) mm, 3 μ m or equivalent
Mobile Phase A	:	0.1percent phosphoric acid in water
Mobile Phase B	:	100 percent acetonitrile
Flow rate	:	1.5 ml/min
Detection	:	UV at 200 nm
Injection	:	10 μ l
Temperature	:	40 °C
Run time	:	14 minutes

Amendment No. 1 to IS 17117 : 2019

GRADIENT TABLE

<i>Sl No.</i>	<i>Time (min)</i>	<i>Mobile Phase A</i> <i>0.1 percent H₃PO₄ in</i> <i>Water</i>	<i>Mobile Phase B</i> <i>100 percent Acetonitrile</i>
(1)	(2)	(3)	(4)
i)	0.0	50	50
ii)	7.0	40	60
iii)	10.0	40	60
iv)	11.0	50	50
v)	14.0	50	50

NOTE — The above gradient table may be modified to ensure specificity and resolution is achieved.

The retention times of the standards are:

<i>Sample No.</i>	<i>Analyte</i>	<i>Xbridge C18 Column</i> <i>Retention Time (min)</i>	<i>YMC Pack ODS Column</i> <i>Retention Time (min)</i>
(1)	(2)	(3)	(4)
1	C12- Betaine	2.19	2.67
2	C12-Glycinate	5.36	7.50

NOTE — The retention times are given for reference and may vary with change in equipment.

H-10 CALCULATION

Determine the weight percent of the surfactant from the calibration curve as follows:

$$\text{Weight percent} = \frac{(\text{Peak area of sample} - \text{intercept of calibration curve})(\text{final sample volume in ml})}{(\text{Slope of calibration curve})(\text{sample weight, in mg})} \times 100 \%$$

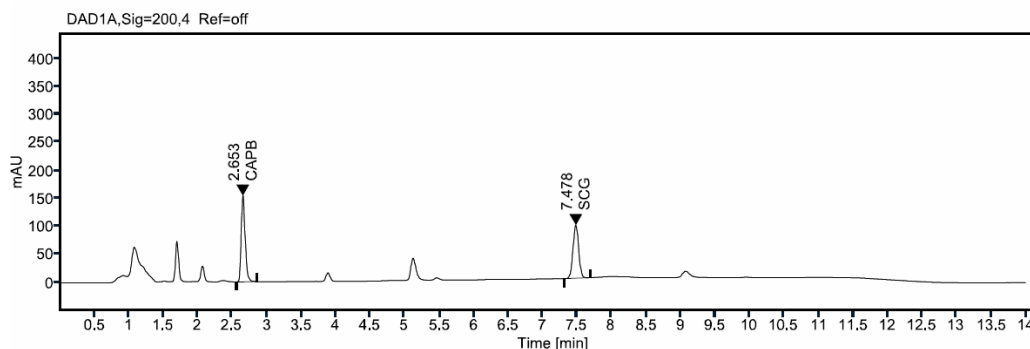
NOTE — Calculations based solely on the C12 peak assume equivalent chain length distributions of the raw material reference standard and the raw material ingredient in the product.

Determination of the weight percent of the surfactant when single standard used in the analysis:

$$\text{Percent content} = \frac{\text{Areas of sample} \times \text{dilution factor of standard} \times \text{purity of standard}}{\text{Average area of standard} \times \text{Dilution factor of sample}}$$

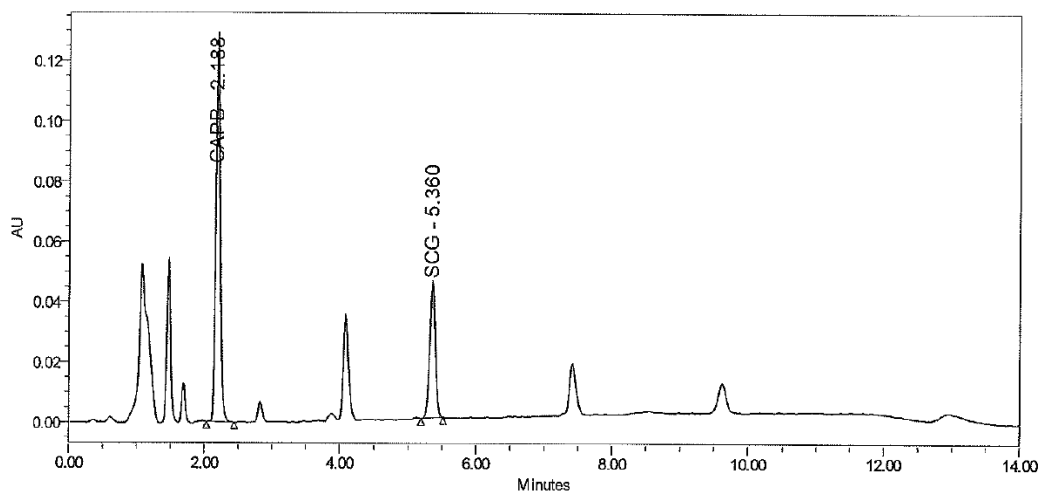
$$\text{Dilution factor of standard/sample} = \frac{\text{Weight of standard or sample}}{\text{Dilution}}$$

H-11 CHROMATOGRAM



Amendment No. 1 to IS 17117 : 2019

Representative chromatogram of a 0.20 mg/ml coco amido propyl betaine and sodium cocoyl glycinate mixed reference standard on YMC pack ODS column



Representative chromatogram of a 0.40 mg/ml coco amido propyl betaine and sodium cocoyl glycinate mixed reference standard on XBridge C18 column

ANNEX J

(Table 1)

DETERMINATION OF FORMALDEHYDE BY UV-VIS METHOD

J-1 SCOPE AND PURPOSE

The purpose of this method is determination of formaldehyde in hair shampoo formulations by UV-VIS spectroscopy.

J-2 SAFETY PRECAUTIONS

The analyst should follow general laboratory practices and use appropriate personal protective equipment. The analyst should read material data sheets (MSDS) of formaldehyde before handling.

J-3 PRINCIPLE OF METHOD

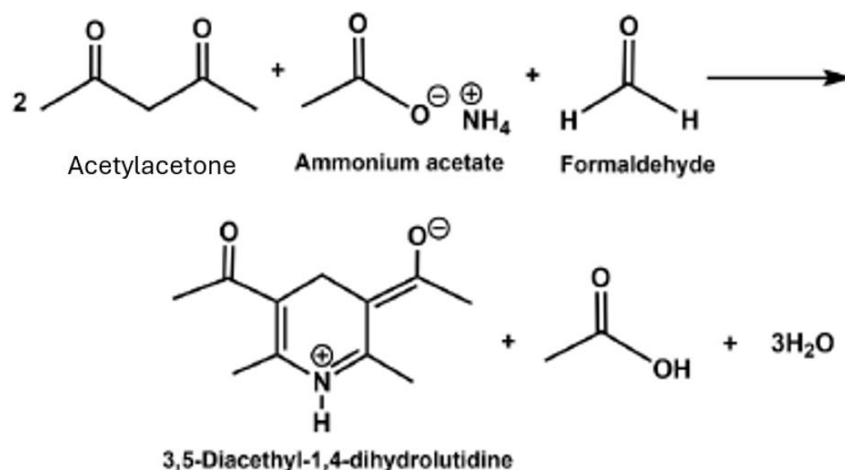
Formaldehyde is reacted with acetyl acetone in an ammonium acetate buffer system to produce a yellow colour derivative. The absorbance of the yellow solution is determined using an ultraviolet-visible spectrophotometer and quantitation is achieved by comparison with a standard.

J-4 FUNDAMENTAL EQUATIONS

Fundamental equation to be added as below:

The formaldehyde concentration was quantified using a Nash reagent composed of acetylacetone and acetic acid/ammonium acetate buffer. Formaldehyde is reacted with an ammonium acetate and acetylacetone under neutral conditions to produce 3,5-diacetyl-1,4-dihydrolutidine as shown in Fig. 1. 3,5-diacetyl-1,4-dihydrolutidine is a yellow compound which is quantitatively calculated with comparison with standard.

Amendment No. 1 to IS 17117 : 2019



J-5 INTERFERENCES

Any extraneous component having a significant absorbance at 412 nm will interfere.

J-6 EQUIPMENT

J-6.1 UV-Visible Spectrophotometer

J-6.2 Quartz Or Optical Glass Cuvette — 1 cm

J-6.3 Water Bath

J-6.4 Centrifuge Tubes — 50 ml

J-6.5 Test Tube — 15 ml, with cap

J-6.6 Micro Pipet — 1 000 μl

J-6.7 Volumetric Flasks — 100 ml and 50 ml

J-6.8 Volumetric Pipettes — 5 ml, 10 ml and 20 ml

J-7. REAGENTS

J-7.1 Formaldehyde Reference Standard — purity ≥ 37 percent, for example, Sigma-Aldrich

J-7.2 Acetyl Acetone (2, 4-Pentanedione) — J. T. Baker Cat# JTS926-6 or equivalent

J-7.3 Acetic Acid — Glacial, Reagent grade

J-7.4 Ammonium Acetate — Reagent grade

J-7.5 Sodium Sulfate — Reagent grade

J-7.6 Isopropanol (IPA) — Reagent grade

J-7.7 Purified Water or equivalent

J-7.8 Acetyl Acetone Working Solution — Weigh 37.5 g of ammonium acetate and quantitatively transfer to 250 ml volumetric flask with purified water. Pipet 0.75 ml of acetic acid and 0.5ml of acetyl acetone into the flask. Dissolve and dilute to the mark with purified water. Mix well, protect from light. This solution should be made fresh for use.

Amendment No. 1 to IS 17117 : 2019

J-7.9 25 Percent Sodium Sulfate Solution — 125g of sodium sulfate transfer gradually to 500 ml volumetric flask which is half filled with purified water, keep mixing with the stirrer. Dissolve and dilute to the mark with purified water, mix well until completely dissolved.

J-8 STANDARD PREPARATION

J-8.1 Standard Stock Solution (Formaldehyde 1 000 µg/ml)

Accurately weigh (0.270 ± 0.01) g of formaldehyde standard (considering 37 percent (w/v) formaldehyde standard) into a 100 ml volumetric flask. Dilute to volume with water and mix well.

J-8.2 Intermediate standard solution (Formaldehyde 100 µg/ml)

Pipette 10.0 ml of standard stock solution into a 100 ml volumetric flask. Dilute to volume with water and mix well.

J-8.3 Working standard Solution: (Formaldehyde 5 µg/mL)

Pipette 5.0 ml of intermediate standard solution into a 100 ml volumetric flask. Dilute to volume with water and mix well.

J-8.4 Prepare another standard solution called as the “Check Standard” by repeating steps J-8.1 through J-8.3.

J-9. SAMPLE PREPARATION

Weigh accurately (1.0 ± 0.001) g into 50 ml centrifuge tube, pipet 20 ml of Sodium Sulfate solution and 20 ml of purified water. Shake well, then keep the sample in a 40 °C bath for 30 minutes. Centrifuge the sample tube for 20 minutes at 3500 rpm. Use supernatant sample solution for derivatization as per 10.1.

J-10 PROCEDURE

NOTE: The blank, standard, and sample solution are placed in the water bath at 40°C and then allowed to react at the same time.

J-10.1 Pipette 5 ml of 5 ppm standard (working standard solution and check standard Solution given in 8.3 and 8.4) and sample (9) into each 15 ml test tube with cap.

J-10.2 Prepare blank by pipetting 5 ml of water (7.7) into a tube.

J-10.3 Pipette 5 ml of acetyl acetone working solution (7.8) into each of the tubes, cap them and mix well.

J-10.4 Equilibrate the tubes in a 40 °C water bath for 30 minutes.

J-10.5 Remove the tubes from the bath, cool to room temperature and add 1ml of Isopropanol to sample tube. If this solution is clear, then add same amount of Isopropanol to blank, standards and additional samples, if any.

J-10.6 If sample is still hazy after 1 ml of IPA added to the sample, add 1 ml of IPA more to standard, sample and blank. Do not add more than 3 ml of total IPA. Mix the solution gently. The solution in each tube must be clear.

Visually check the tube containing sample solution for the development of yellow color and measure the absorbance at 412 nm wavelength as per section 11.

NOTE — For sample that are highly coloured, the sample can be analysed without acetyl acetone to check the possible interference from the sample matrix or as appropriate.

J-10.7 LOQ and LOD Value

LOD = 0.125 ppm

LOQ = 0.5ppm

Amendment No. 1 to IS 17117 : 2019

J-11 OPERATING CONDITIONS

J-11.1 Turn on the lamp of the UV/Visible spectrophotometer.

J-11.2 Warm up the lamp for at least 15 minutes.

J-11.3 Set the instrument at a fixed wavelength of 412 nm.

J-11.4 Autozero the instrument with blank solution.

J-11.5 Use the blank solution as reference for all further measurements.

J-11.6 Rinse the cuvette carefully with the Working Standard Solution and fill it to the appropriate level.

J-11.7 Insert the cuvette into the cuvette holder. The cuvette needs to be inserted the same way each time during the readings.

J-11.8 Read the absorbance for the working standard solution

J-11.9 Read absorbance for the check standard solution. Calculate the percentage of check standard conformity. The percentage of check standard conformity must agree within 97.0 percent to 103.0 percent.

NOTE — If the percentage of check standard conformity does not agree within 97.0 percent to 103.0 percent, the third standard should be prepared according to section 8 (standard preparation). Read the absorbance for the third standard. The percentage of check standard conformity must agree within 97.0 percent to 103.0 percent considering the third standard.

J-11.10 Read the absorbance for the sample solution.

J-11.11 Read the absorbance of the reference standard once after maximum of ten sample readings and at the end of the run.

J-11.12 The percent RSD for absorbance of the complete set of the reference standard measurements should not exceed 4.0 percent.

J-11.13 Use the average absorbance of all the working standard solutions (STD) for sample calculation.

J-12 CALCULATION AND REPORTING

J-12.1 Check standard conformity:

$$\text{Percent check standard conformity} = \frac{A_2 \times W_1 \times 100}{A_1 \times W_2}$$

where

A_2 = absorbance reading of check standard;

W_1 = mass, in g, of working standard;

A_1 = absorbance reading of working standard; and

W_2 = mass, in g, of check standard.

J-12.2 The concentration of formaldehyde in sample:

$$\text{ppm of formaldehyde} = \frac{\text{Absorbance of sample} \times \text{ppm of STD} \times 40}{\text{Average absorbance of STD} \times \text{weight of sample}}$$

where

40 = dilution factor;

Weight of sample = mass, in g, of sample; and

STD = working standard solution.

$$\begin{aligned} \text{ppm of STD} &= \frac{X \text{ g} \times 10 \times 5 \times \text{percent purity of formaldehyde} \times 10\,00\,000}{100 \times 100 \times 100 \times 100} \\ &= X \text{ g} \times \text{percent purity of formaldehyde} \times 0.5 \end{aligned}$$

ANNEX K

(Table 1)

DETERMINATION OF FORMALDEHYDE BY HPLC METHOD

K-1 OBJECTIVE OF THE METHOD

Quantification of formaldehyde in finished products by derivation with 2,4-dinitrophenylhydrazine (2,4-DNP) then reverse phase assay on a column type C18.

The quantification of formaldehyde is recommended by dosed additions. Indeed, accuracy and precision are better than quantification by external calibration. With dosed additions, all samples are derived under the same conditions by being integrated into the same bypass reactor. Screening of aldehydes in finished products.

K-2 PRINCIPLE

Reverse phase HPLC assay after reaction medium derivation of formaldehyde by DNPH by UV/visible detection. Qualitative screening of aldehydes in finished products by this same methodology.

K-3 MATERIALS REQUIRED

K-3.1 Hardware

K-3.1.1 HPLC — with diode array detector or equivalent

K-3.1.2 HPLC — C18 column; 250 × 4.6, 5µm

K-3.1.3 pH Meter — Mettler Toledo or equivalent

K-3.1.4 Thermomixer Comfort Eppendorf

K-3.1.5 support Flacon — 15 ml Falcon 15 ml

K-3.1.6 Polypropylene Reaction Tube

K-3.1.7 Filter — ACRODISC GHP 25 mm 0.45µm

K-3.2 Reagents and Scales

K-3.2.1 Formaldehyde — (CAS No. 50-00-0) in 16 percent aqueous solution (m/v) in 10 ml ampoule

K-3.2.2 2,4-Dinitrophenylhydrazine (DNPH) — (CAS No. 119-26-6) 97 percent stabilized with 30 percent water

K-3.2.3 37 percent Hydrochloric Acid (CAS No. 7647-01-0)

K-3.2.4 Acetonitrile (ACN) (CAS No. 75-05-8) — Gradient Grade

K-3.2.5 Water Quality HPLC

K-4 METHODOLOGY

K-4.1 Preparation of Reagents

Preparation of a derivation solution based on DNPH at 1.5 g/l:

- Introduce 310 mg of 97 percent DNPH in a 200 ml volumetric flask;
- Complete with the gauge line with the acetonitrile gradient grade; and
- This solution is kept at 4 °C away from light no more than two days.

Amendment No. 1 to IS 17117 : 2019

K-4.2 Analytical Conditions

- a) Chromatographic conditions for quantification by external calibration
- Track A: ACN B: MeOH
Track C: ACN D: Water quality HPLC
- b) Injection of 20 µl.
- c) Detection: (recording from 320 nm to 400 nm) 360 nm and 349 nm column oven thermostated at 30 °C.
- d) Sample compartment at 6 °C.
- e) For information, at T0 the pressure is about 1650 psi.
- f) For the determination of formaldehyde.

The analysis is first done in isocratic mode and then in a second time in gradient mode to eliminate all other unanalyzed components from finished products.

	<i>Time</i>	<i>Flow</i>	<i>% A</i>	<i>% B</i>	<i>% C</i>	<i>% D</i>	<i>Curve</i>
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	0.01	1.20	0.0	0.0	60.0	40.0	6
2	6.00	1.20	0.0	0.0	60.0	40.0	6
3	13.00	1.20	0.0	0.0	69.0	31.0	6
4	14.00	1.20	0.0	0.0	100.0	0.0	6
5	16.00	1.20	0.0	0.0	100.0	0.0	6
6	16.10	1.20	0.0	0.0	60.0	40.0	1
7	20.00	1.20	0.0	0.0	60.0	40.0	6

For the qualitative screening of aldehydes. The analysis is done first in isocratic mode and then in a second time in gradient mode to eliminate all other unanalyzed components from the finished products.

	<i>Time</i>	<i>Flow</i>	<i>% A</i>	<i>% B</i>	<i>% C</i>	<i>% D</i>	<i>Curve</i>
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	0.01	1.20	0.0	0.0	60.0	40.0	6
2	6.00	1.20	0.0	0.0	60.0	40.0	6
3	13.00	1.20	0.0	0.0	69.0	31.0	6
4	14.00	1.20	0.0	0.0	100.0	0.0	6
5	30.00	1.20	0.0	0.0	100.0	0.0	6
6	31.00	1.20	0.0	0.0	60.0	40.0	6
7	40.00	1.20	0.0	0.0	60.0	40.0	6

K-4.3 Measurements, retention time

These retention times are given as an indication, under the conditions of the test.
HPLC visible detection, $\lambda = 353$ nm for: Formaldehyde at around 4.20 min.

K-4.4 Preparation of standards/calibration range

K-4.4.1 Preparation of Standards

Preparation for external calibration: Formaldehyde

Amendment No. 1 to IS 17117 : 2019

K-4.4.2 Range of the Aldehyde

To prepare the mix 3 mol stock solution and the diluted solution, a standard range is prepared in 20 ml vials, not forgetting to add in each vial 30 µl of 37 percent hydrochloric acid for the diversion which must be done in an acidic medium.

<i>Sl No.</i>		<i>Stock Sol Mix 3 mol (ml)</i>	<i>Diluted Sol Mix 3 mol (ml)</i>	<i>HCl at 37 percent (ml)</i>	<i>Ampoules (ml)</i>	<i>Concentration Final Formaldehyde (µg/ml)</i>
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	White	-	-	0.030	QSP 20 ml H ₂ O	0.00
ii)	STD 1	-	0.100	0.030	QSP 20 ml H ₂ O	0.200
iii)	STD 2	-	0.200	0.030	QSP 20 ml H ₂ O	0.400
iv)	STD 3	-	0.500	0.030	QSP 20 ml H ₂ O	1.00
v)	STD 4	-	1.000	0.030	QSP 20 ml H ₂ O	2.00
vi)	STD 5	0.050	-	0.030	QSP 20 ml H ₂ O	10.0
vii)	STD 6	0.100	-	0.030	QSP 20 ml H ₂ O	20.0
viii)	STD 7	0.200	-	0.030	QSP 20 ml H ₂ O	40.0

NOTE — It is essential to make a blank to estimate the amount of formaldehyde in the DNPH.

K-4.4.3 Derivation of the Range by the 2,4-DNPH (External Calibration)

Standards and white must be derived by the DNPH before being analyzed. The leads are made at 50 °C. Before starting the preparation, it is important to program the thermomixer(s) at the right temperature.

K-4.4.4 Formaldehyde

- Prepare and label 8 ml to 15ml 'Falcon' tubes;
- For each of 8 solutions of the standard range put 5 ml in one of the 8 tubes 'Falcon';
- In each tube add 5 ml of the DNPH bypass solution;
- Homogenize rapidly at the vortex;
- Place the reaction tubes in the already temperature-stabilized thermomixer;
- Cover the vials with aluminum foil. The reaction must be to the dark;
- Program and start the appropriate cycle: From 2H to 50 °C with a stirring of 700 rpm;
- At the end of this time, filter on GHP ACRODISC filter 0.45µm in the injection vials;
- The vials are kept away from light at 6°C in the cooling compartment of the HPLC system before injection; and
- 20 µl of each solution is injected to establish the calibration range. Vials must be kept away from light between 6 °C and 10 °C.

K-4.5 Preparation for External Calibration — Formaldehyde

K-4.5.1 In a 100 ml volumetric flask, weigh about 200 mg of sample (finished product or raw material).

K-4.5.2 Add about 50 ml of water and then add 0.150 ml of 37 percent hydrochloric acid.

K-4.5.3 Complete to the gauge line by water quality HPLC.

Amendment No. 1 to IS 17117 : 2019

K-4.5.4 If the product is presumed to be highly concentrated, this solution may be diluted. This dilution should be done in water quality HPLC acidified by 30 µl of 37 percent hydrochloric acid per 20 ml of solution.

K-4.5.5 The derivation by the DNPH is then carried out in a 'Falcon' tube as indicated below.

K-4.5.6 In a tube:

- a) place 5 ml of the acidified solution finished product and raw material; and
- b) Then add 5 ml of the DNPH bypass solution.

K-4.5.7 Homogenize rapidly at the vortex.

K-4.5.8 Place the reaction tubes in the thermomixer.

K-4.5.9 Cover the vials with aluminum foil. The reaction must be to the dark.

K-4.6 Program and start the appropriate cycle: from 2H to 50 °C with a stirring of 700 rpm.

K-4.6.1 At the end of this time, filter on GHP ACRODISC filter 0.45 µm in the injection vials.

K-4.6.2 The vials are kept away from light at 6 °C in the cooling compartment of the HPLC system before injection.

K-4.6.3 Perform 3 tests per sample.

K-4.6.4 Homogenize rapidly at the vortex.

K-4.6.5 Place the reaction tubes in the thermomixer.

K-4.6.6 Cover the vials with aluminum foil: The reaction should be in the dark. Program and start the cycle of 30 minutes at 10 °C with a stirring of 700 rpm.

K-4.6.7 At the end of this time, filter on GHP ACRODISC filter 0.45 µm in the injection vials.

K-4.6.8 The vials are kept away from light at 6°C in the cooling compartment of the HPLC system before injection.

For External Calibration: Glyoxylic acid, Acetaldehyde, Formaldehyde, Glyoxal

For formaldehyde and acetaldehyde, detection is made at 360 nm. Standard settings allow for good integration. The calculation of the calibration line is carried out on the surface of the chromatographic peaks in linear regression with a weighting in 1/X.

Calculation taking into account the dilution of the samples:

Measure the surface area of the peaks corresponding to the formaldehyde, for the five samples prepared. Draw a straight line by carrying on abscissa the concentrations du compound added in the mixture (matrix + standard) in µg/g and in ordinates the surfaces of the corresponding peak.

We obtain a line of the type: $y = Ax + B$

Injected formaldehyde content (µg/g):

$$C_i = \frac{B}{A}$$

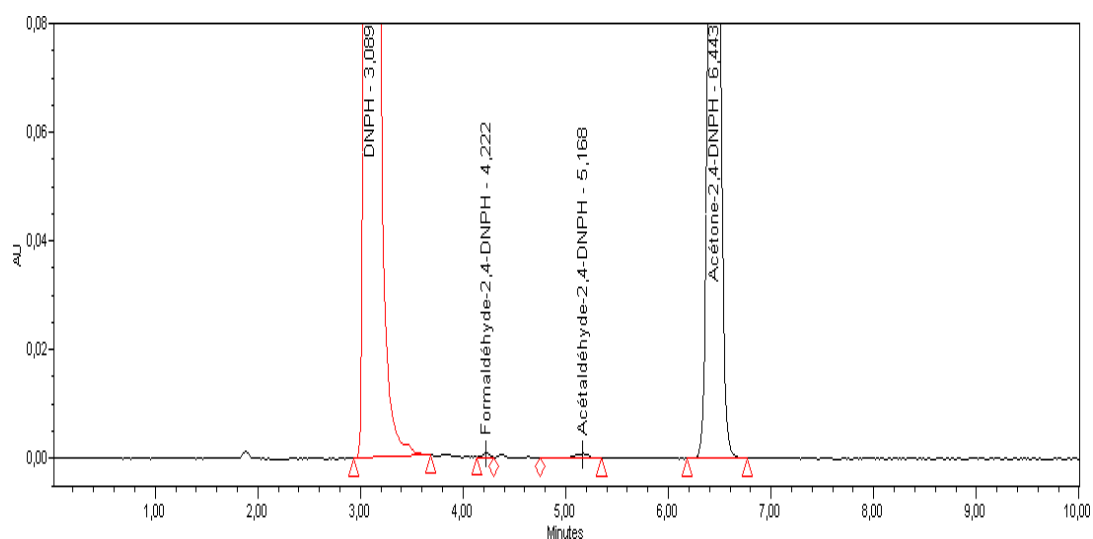
Formaldehyde concentration in the matrix (µg/g sample):

Amendment No. 1 to IS 17117 : 2019

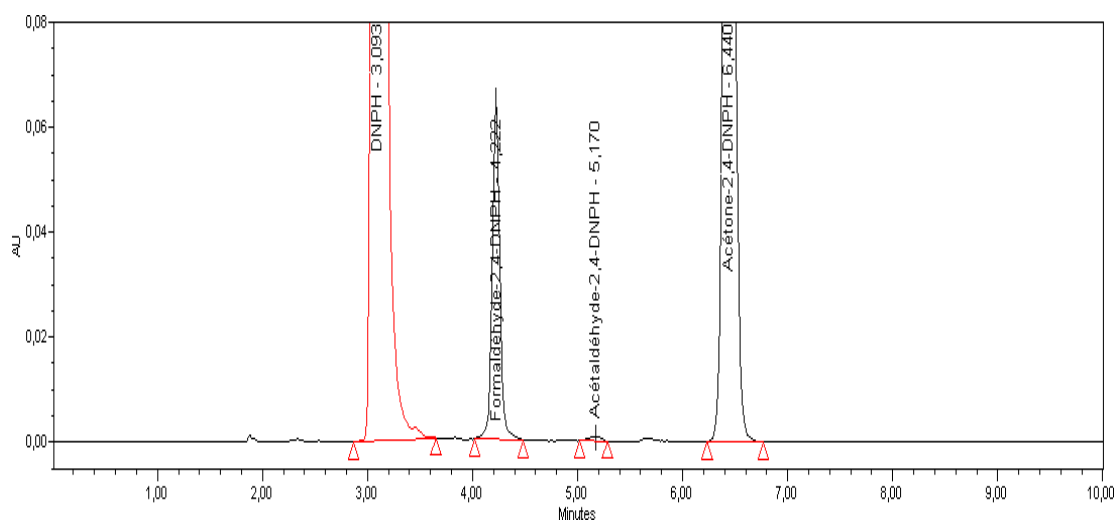
$C_M = C_i \times \frac{M_T}{M_{PE}}$	with: a) A = Slope of the calibration line; b) B = Intercept of the calibration line; c) C_i = Concentration of injected formaldehyde (µg/g); d) C_M = Formaldehyde concentration in the matrix (µg/g sample); e) M_T = Total mass of the prepared control sample ECHO (matrix + solvent) (g); and f) M_{PE} = Mass of the test portion of the control matrix ECHO (g).
---------------------------------------	--

K-5 CHROMATOGRAMS AND SPECTRA: FORMALDEHYDE

Chromatogram of a white DNP

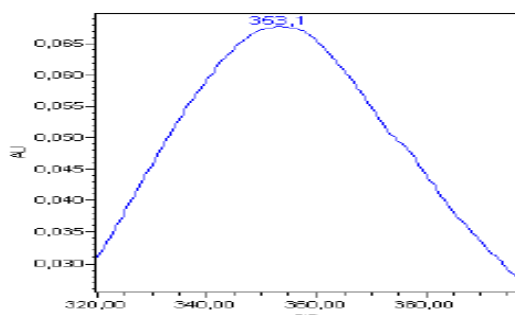


Chromatogram of point 5 of the range



Spectrum of formaldehyde-2,4-DNPH

Amendment No. 1 to IS 17117 : 2019



ANNEX M

(Table 1)

DETERMINATION OF FORMALDEHYDE BY GC HS METHOD

M-1 ANALYSIS REQUIREMENT

M-1.1 Gas chromatography with HS sampler or any other equivalent

M-1.2 GC Column — DB-624 capillary column, 30m length, 0.32mm inner diameter, 1.8 μ m film thickness. (Part. No: -123-1334, Make: - Agilent) or any other equivalent

M-1.3 Working Standard/Reference Standard — Formaldehyde solution 37 percent (sigma Aldrich)

M-1.4 Sample Details — Shampoo

M-1.5 Chemical/Reagent Details:

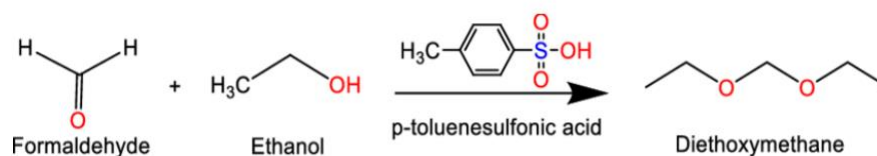
Ethanol (AR)

p-Toluene sulphonic acid monohydrate (PTSA) (AR) (CAS No. 6192-52-5)

M-1.6 Principle and Reaction and Representative Chromatogram

Principle and reaction and representative chromatogram to be included which help analyst to add value while doing and learning.

Reaction:



M-2 CHROMATOGRAPHIC CONDITION

Column	:	DB-624 capillary column, 30m length, 0.32 mm inner diameter, 1.8 μ m Film thickness. (Part. No: -123-1334, Make: - Agilent) or any other equivalent
Detector	:	FID at 300 $^{\circ}$ C
Mobile phase	:	Nitrogen gas
Elution type	:	Temperature gradient (Refer table No. 2)
Flow rate	:	1.0 ml/minute
Diluent	:	1.5 percent PTSA in ethanol (AR grade)
GC cycle time	:	50 minutes

Amendment No. 1 to IS 17117 : 2019

HS Oven temperature : 70 °C
Sample line temperature : 110 °C
Transfer line temperature : 140 °C
Shaking level : 4

Pressurize gas pressure : 100 kPa
Split ratio : 1:5
HS equilibration time : 50 min
Pressurization time : 2.0 min
Pressure equilibration time : 0.1 min
Load time : 0.5 min
Load equilibration time : 0.1 min
Injection time : 1.0 min
Needle flush time : 10 min

M-2.1 Column Oven Temperature Program

<i>Sl No.</i>	<i>Rate/min</i>	<i>Temperature</i>	<i>Hold (min)</i>
(1)	(2)	(3)	(4)
i)	-	35 °C	7
ii)	5°C/min	120 °C	1
iii)	50°C/min	220 °C	9

M-2.2 Preparation of 1 Percent para-Toluene Sulphonic Acid in Ethanol

Accurately weigh 7.5 g of para-toluene sulphonic acid monohydrate and dissolve in 500 ml absolute ethanol.

M-2.3 Standard Preparation (Stock Solution)

Accurately weigh 135 mg of 37 percent formaldehyde standard solution to a 50 ml volumetric flask, add about 30 ml diluent, sonicate for 2 min. Make the volume up to the mark with diluent and mix well.

M-2.4 Working Standard Solution

Pipette out 1 ml of the stock solution to a 100 ml volumetric flask and dilute up to the mark with diluent. Pipette out 5 ml of the working standard solution into a standard GC-headspace vial and seal it with aluminum crimp cap with silicone septa.

M-2.5 Sample Preparation

Accurately weigh 250 mg of sample to a standard 20 ml GC-headspace vial and add 5 ml diluent. Seal the HS-vial with aluminum crimp cap with silicone septa.

M-2.6 System Suitability

Inject the diluent as blank until a stable baseline is achieved. Inject the standard and again inject the blank to check the carryover.

Make six replicate injections of standard solution and check the percent RSD of the area of standard. Continue the sample analysis once the following system suitability parameters are achieved.

Retention time for formaldehyde	15 minute
Relative standard deviation (percent RSD for 6 replicate injections)	NMT10.0 percent
Tailing factor	NMT2.0
No. of theoretical plates	NLT2000

Amendment No. 1 to IS 17117 : 2019

M-3 CALCULATION

$$FC = \{[(A_1)/(A_2)] \times [(W_2)/(50)] \times [(1)/(100)] \times [(5)/(W_1)] \times (P) \times (10000)\}$$

where

A_1 = average area of sample peak corresponding to standard peak;

A_2 = average area of standard peak;

W_1 = mass of sample (mg);

W_2 = mass of standard (mg);

P = percent purity of standard; and

FC = formaldehyde content (ppm).

(PCD 19)